



Universität Stuttgart

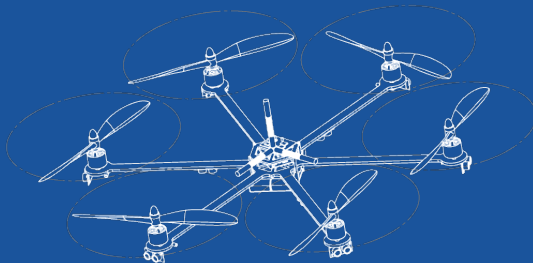
Institut für Flugmechanik und Flugregelung

Region of attraction with integral quadratic constraints

System Theoretical Methods for Flight Control

Saturday 16th May, 2026

First Last



- ① Introduction
- ② Traditional Approach
- ③ Limitation
- ④ Proposed Solution
- ⑤ Key Advantage
- ⑥ Implementation
- ⑦ Validation
- ⑧ TikZ Examples
- ⑨ Useful Features

Today's Agenda

① Introduction

Goal of the paper is to design a framework for safe control design from experimental data.
The system is given as:

$$\dot{\mathbf{x}}(t) = f(\mathbf{x}) + g(\mathbf{x})u(t) + \mathbf{w}(t)$$

Safe control design from experimental data

Today's Agenda

① Introduction

② Traditional Approach

Barrier functions for safety certification

Today's Agenda

- ① Introduction
- ② Traditional Approach
- ③ Limitation

Non-convex optimization, computational challenges

Today's Agenda

- ① Introduction
- ② Traditional Approach
- ③ Limitation
- ④ Proposed Solution

Density functions for data-driven control

Today's Agenda

- ① Introduction
- ② Traditional Approach
- ③ Limitation
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- ⑤ Key Advantage

Convex reformulation using semidefinite programming (SDP)

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- ⑥ Implementation

Casos tool for controller and observer synthesis

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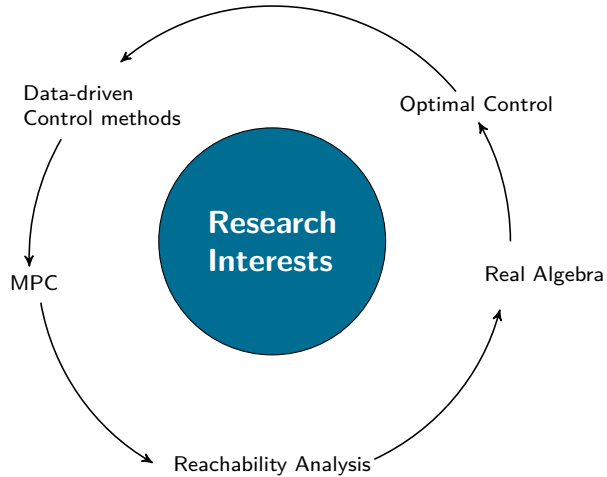
Experimental verification and performance analysis

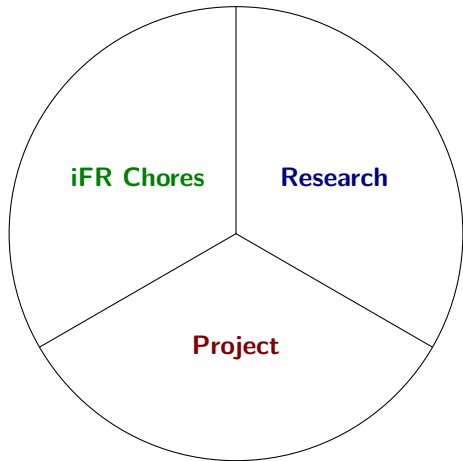
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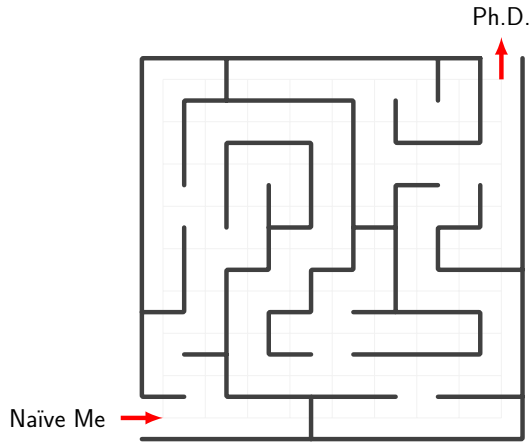


In the next slides, we will see some examples.





TikZ Example - 3



TikZ Example - 4

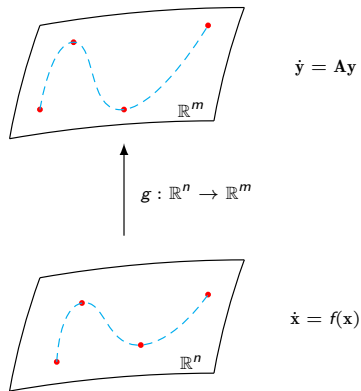


Figure 1: Some description

Well, tikz is AWESOME, however as you probably noticed by now, it can take a lot of time compiling the whole document if it contains multiple tikzpictures.

If possible, generate the tikzpictures in a standalone file, i.e., generate .pdf files with the figures and only import those to the main file. Believe me, it will reduce by a lot the time you spend waiting for the compiled file.

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Since this is latex, we can use equations.

$$\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu} \tag{1}$$

All equations are numbered by the order they appear in the presentation. But if you don't want a numbered equation, you can still do it

$$E = mc^2$$

One useful and nice feature is the block environment (*newtheorem*). It can be used to highlight important information. (see `preamble.tex`)

Conferences

- ① *American Control Conference, Toronto, July 2024*
Initial deadline: **September 22, 2023**
- ② *European Control Conference, Stockholm, June 2024*
Initial deadline: **October 25, 2023**

Theorem (Theorem name)

This is a theorem.

Remark

This is a remark.

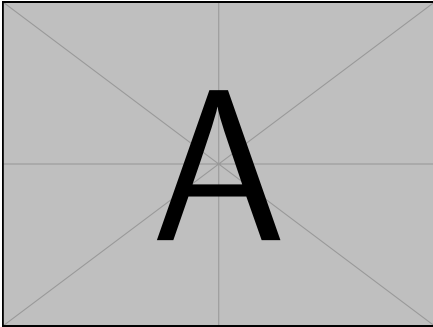


Figure 2: Caption 1

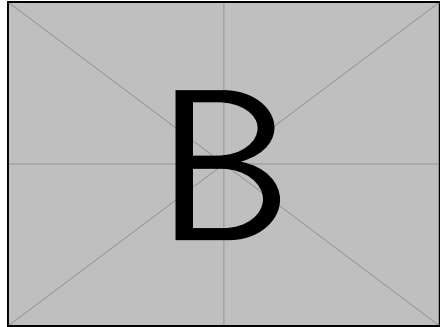
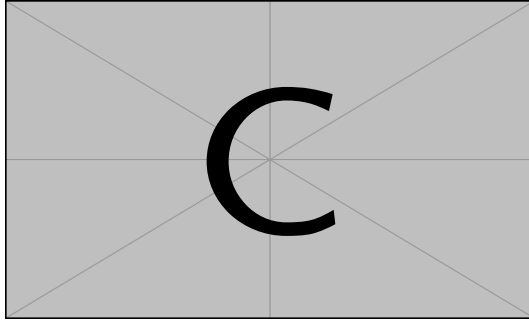


Figure 3: Caption 2



Note that only some pdf viewer can render the video. For presentations I recommend using '*pdfpc*', an open source presentation viewer that mimics the presentation layout from PowerPoint (some advantages include a timer, a way to see the next slide before it is presented and the use of a pointer or a pen)

Just as a normal .tex document we can use .bib files to cite references.

- Some random text... [1]
- Another random text... [2, 3]

The references are shown at the end of the presentation automatically.

In the next slide, you have the default final slide. You can edit it as you see fit.
In the current state you just need to change the photo and a few information details.



Universität Stuttgart

Institute of Flight Mechanics and Controls

Thank you!



First Last

E-Mail: ...@ifr.uni-stuttgart.de

- [1] Russell A. Smith. Some Applications of Hausdorff dimension inequalities for ordinary differential equations. *Proc. Royal Society of Edinburgh: Section A Mathematics*, 104(3-4):235–259, 1986.
- [2] Pablo A. Parrilo. Semidefinite programming relaxations for semialgebraic problems. *Mathematical Programming, Series B*, 96(2):293–320, 2003.
- [3] James S. Muldowney. Compound matrices and ordinary differential equations. *Rocky Mountain Journal of Mathematics*, 20(4):857–872, 1990.